

Rentals KPI Formula Sheet

EstateCFO - Overview, Financials, CFO Dashboard, Financial Ratios, AR Dashboard

Rentals KPI Formula Sheet

Reference for KPI calculations across Rental Overview, Financials KPI Dashboard, CFO Dashboard, and CFO Portfolio.

A) Rental Overview (operational dashboard)

Data source: unit registry, invoices, collections, expenses, QB AR Aging upload

Card Label	Formula
Occupancy Rate	Occupied Units / Total Units x 100
Occupied / Vacant	Occupied Units and Vacant Units (vacant = total - occupied)
Collected This Month	Sum of collections where collected_date = selected month
NOI This Month	Collected This Month - Total Expense This Month
Gross Potential Rent	Sum of all unit monthly_rent
Vacancy Loss	Sum of monthly_rent for units with status = vacant only
Collection Rate	Collected This Month / Billed This Month x 100
Avg Rent / Unit	Gross Potential Rent / Occupied Units
Arrears Days Outstanding	Weighted DSO from QB aging buckets: (Currentx0 + 1-30x15 + 31-60x45 + 61-90x75 + 90+x105) / Total
Vacant > 30 Days	NA (vacancy date not tracked yet)
NOI Margin	NOI This Month / Collected This Month x 100
Best / Worst (Occ.)	Highest and lowest company occupancy %
Arrears Outstanding	Sum of max(0, Billed - Collected) per invoice
Partner Share Payable	NOI x ownership % for limited/silent partners (only if NOI > 0)

AR section (Overview)

Item	Formula
Arrears Aging buckets	Unpaid amount per invoice aged from billing month due date (1st of month) into Current / 1-30 / 31-60 / 61-90 / 90+
Top Risk Companies	Ranked by Arrears + Vacancy Loss; flagged if arrears > \$2k, occupancy < 85%, or high arrears
Company filter	Uses per-company QB aging totals when uploaded

Backend reference: backend/services/rental_calculations.py (company_summary, arrears_aging, unit_arrears)

B) Financials -> KPI Dashboard (P&L / Balance Sheet)

Data source: uploaded QuickBooks financials (CASH_FLOWS.xlsx) Frontend reference: frontend/src/utils/rentalKpiEngine.ts, frontend/src/pages/RentalFinancials.tsx (KPITab)

Base inputs (from P&L / BS)

Input	Formula
Total Revenue	Total Income row, else sum of income/revenue/rent lines
Total Expenses	Total Expenses row
Net Income	Net Income row
NOI	Net Operating Income row, else Total Revenue - Total Expenses + Interest Expense
Interest Expense	Interest paid lines (absolute value)
Rental Income	Rental income / services lines
Management Fee	Management fee lines
Repairs	Repair / maintenance / cleaning lines
Buildings	Buildings / property & equipment from balance sheet
Long-term Loans	Long-term liabilities / loan accounts
Cash	Total bank accounts

Input	Formula
Total Assets / Liabilities / Equity	Balance sheet totals

Profitability KPIs

Card Label	Formula	Target
NOI Margin	$\text{NOI} / \text{Total Revenue} \times 100$	> 40%
Net Income Margin	$\text{Net Income} / \text{Total Revenue} \times 100$	> 10%
Revenue Growth YoY	$(\text{Current Revenue} - \text{Prior Revenue}) / \text{Prior Revenue} \times 100$	> 3%
Expense Ratio	$\text{Total Expenses} / \text{Total Revenue} \times 100$	< 70%

Rental performance KPIs

Card Label	Formula	Target
Rental Income %	$\text{Rental Income} / \text{Total Revenue} \times 100$	> 80%
Interest Coverage	$\text{NOI} / \text{Interest Expense}$	> 2.0x
Mgmt Fee %	$\text{Management Fee} / \text{Total Revenue} \times 100$	< 10%
Repair % of Revenue	$\text{Repairs} / \text{Total Revenue} \times 100$	< 5%

Balance sheet KPIs

Card Label	Formula	Target
LTV (Loans / Building)	$\text{Long-term Loans} / \text{Buildings} \times 100$	< 75%
Asset / Liability Ratio	$\text{Total Assets} / \text{Total Liabilities}$	> 1.5x
Debt-to-Equity	$\text{Total Liabilities} / \text{Equity}$	< 2.0x
Cash Balance	Bank account total from balance sheet	> \$10K
Debt-to-Asset %	$\text{Total Liabilities} / \text{Total Assets} \times 100$	< 80%
Equity Ratio %	$\text{Equity} / \text{Total Assets} \times 100$	> 20%
Net Debt	Long-term Loans - Cash	Monitor
DSCR (Est.)	$\text{NOI} / (\text{Interest Expense} \times 1.2)$	> 1.25x

Occupancy KPIs (when ops data is linked)

Card Label	Formula	Target
Occupancy Rate	From registry occupancy %	> 95%
Economic Occupancy / Rent Collection Rate	$\text{Collected} / \text{Billed} \times 100$	> 95%
Vacancy Rate	$100 - \text{Occupancy Rate}$	< 5%
Cap Rate	$\text{NOI} / \text{Buildings} \times 100$	> 5%
Revenue per Unit	$\text{Total Revenue} / \text{Total Units}$	Trend
Expense per Unit	$\text{Total Expenses} / \text{Total Units}$	Trend

C) CFO Dashboard (Financial Overview - P&L charts)

Data source: first company's uploaded financials Route: Rentals -> CFO Dashboard Frontend reference: frontend/src/pages/RentalCfoDashboard.tsx

Metric / Chart	Formula		
Revenue	Total Revenue or sum of revenue/rental income lines		
Expenses	Sum of expense lines		
Net Income	Net Income row		
Net Margin	$\text{Net Income} / \text{Revenue} \times 100$		
Expense Ratio Trend	Expenses / Revenue x 100 by year		
Cash (Bank)	Total Bank Accounts from balance sheet		
Net Income Trajectory	Net Income by year		
Latest Net Income change	$(\text{Latest NI} - \text{Prior NI}) / \text{Prior NI}$	Prior NI	x 100`
Avg Profit Margin	Average of yearly (Net Income / Revenue x 100)		

D) CFO Portfolio (strategic portfolio view)

Data source: companies, expenses, units, loans, portfolio summary Route: Rentals -> CFO Portfolio Frontend reference:

Top KPI tiles

Card Label	Formula
NOI	Collected This Month - Total Expenses (filtered companies)
Occupancy	Occupied Units / Total Units
Collection Rate	Collected / Billed
Expense Ratio	Total Expenses / Gross Potential Rent
DSCR	$(\text{NOI} \times 12) / (\text{Total Monthly EMI} \times 12)$
Cash Position	NOI - Total Monthly EMI
Rent Growth	Currently static placeholder (+2.4%)
Vacancy Cost	Portfolio vacancy_loss

Portfolio Health Score (out of 100)

Component	Max Points	Formula
Occupancy	25	$\min(25, \text{Occupancy \%} \times 25)$
Collections	25	$\min(25, \text{Collection Rate} \times 25)$; if no billing data -> 15
Expense Control	25	25 if expense ratio < 30%; 18 if < 45%; else 10
Debt Coverage	25	25 if DSCR > 1.25; 18 if >= 1.0; 12 if no loan data; else 8

Total Health Score = sum of the 4 components (averaged across filtered companies)

Company Comparison Matrix

Column	Formula
Occupancy	Company occupancy %
NOI	Collected - Expenses for the month
Exp Ratio	Expenses / Gross Potential Rent
DSCR	$(\text{NOI} \times 12) / (\text{Company EMI} \times 12)$
Cash	NOI - Company EMI
Score	Same health score formula above
Flag	Score < 70 or DSCR < 1.0

Building-level (Building Expenses view)

Metric	Formula
Rent Income	Collected (or GPR if no collection)
Total Expenses	Sum of building expenses for current month
NOI	Rent Income - Total Expenses
Expense Ratio	Total Expenses / Rent Income
NOI Margin	NOI / Rent Income
Status	Healthy if ratio < 30%; Watch if <= 45%; High if > 45%

G) Financial Ratios page

Route: Rentals -> Financial Ratios Frontend reference: frontend/src/pages/rental/RentalFinancialRatios.tsx

Profitability ratios

Ratio	Formula	Benchmark
NOI Margin	$\text{NOI} / \text{Revenue} \times 100$	> 35%
Net Profit Margin	$\text{Net Income} / \text{Revenue} \times 100$	> 10%
Operating Expense Ratio	$\text{Total OpEx} / \text{Revenue} \times 100$	< 60%
EBITDA Margin	$\text{EBITDA} / \text{Revenue} \times 100$	> 45%
Return on Assets (ROA)	$\text{Net Income} / \text{Total Assets} \times 100$	> 4%

Ratio	Formula	Benchmark
Return on Equity (ROE)	$\text{Net Income} / \text{Equity} \times 100$	> 8%
Gross Rent Multiple	$\text{Asset Value} / \text{Annual Revenue}$	< 14x
Cash-on-Cash Return	$\text{Pre-tax Cash Flow} / \text{Equity} \times 100$	> 7%

Where: - NOI = Revenue - Expenses + Interest Expense - EBITDA = NOI + Depreciation + Amortization

Liquidity ratios

Ratio	Formula	Benchmark
Current Ratio	$\text{Current Assets} / \text{Current Liabilities}$	> 1.5x
Quick Ratio	$(\text{Current Assets} - \text{Inventory}) / \text{Current Liabilities}$	> 1.0x
Cash Ratio	$\text{Cash \& Bank} / \text{Current Liabilities}$	> 0.2x
Operating CF Ratio	$\text{Operating Cash Flow} / \text{Current Liabilities}$	> 1.0x
Working Capital	$\text{Current Assets} - \text{Current Liabilities}$	Positive
Days Cash on Hand	$\text{Cash} / (\text{Annual OpEx} / 365)$	> 60 days

Solvency / leverage ratios

Ratio	Formula	Benchmark
Debt-to-Equity	$\text{Total Liabilities} / \text{Equity}$	< 5x (RE)
Debt-to-Asset	$\text{Total Liabilities} / \text{Total Assets} \times 100$	< 80%
Equity Ratio	$\text{Equity} / \text{Total Assets} \times 100$	> 20%
Interest Coverage	$\text{NOI} / \text{Interest Expense}$	> 1.5x
LTV	$\text{Mortgage} / \text{Long-term Loans} / \text{Property Value (Buildings)} \times 100$	< 80%
Net Debt	$\text{Long-term Loans} - \text{Cash}$	Monitor
DSCR	$\text{NOI} / \text{Total Debt Service (est: NOI} / \text{Interest} \times 1.2)$	> 1.25x
Debt Service Ratio	$\text{Debt Service} / \text{NOI} \times 100$	< 65%
Fixed Charge Coverage	$(\text{NOI} + \text{Fixed Charges}) / \text{Fixed Charges}$	> 1.25x

Occupancy & pricing ratios

Ratio	Formula	Benchmark
Occupancy Rate	$\text{Occupied Units} / \text{Total Units} \times 100$	> 90%
Economic Occupancy	$\text{Rent Collected} / \text{Gross Potential} \times 100$	> 92%
Rent Collection Rate	$\text{Collected} / \text{Billed} \times 100$	> 95%
Vacancy Rate	$\text{Vacant Units} / \text{Total Units} \times 100$	< 10%
Loss to Lease	$(\text{Market Rent} - \text{Actual Rent}) / \text{Market Rent} \times 100$	< 5%
Avg Days Vacant	$\text{Average days between tenants}$	< 21–30 days
Rent per Sq Ft	$\text{Average Rent} / \text{Average Sq Ft}$	Market
Revenue per Unit	$\text{Total Revenue} / \text{Total Units}$	Trend
Expense per Unit	$\text{Total OpEx} / \text{Occupied Units}$	< \$1,000/mo
Cap Rate	$\text{NOI} / \text{Property Value} \times 100$	4–6% (market)
Price / Rent Ratio	$\text{Property Value} / \text{Annual Rent}$	< 20x
EGIM	$\text{Property Value} / \text{Effective Gross Income}$	< 12x

Returns / cost of capital

Ratio	Formula
WACC	Weighted average cost of debt and equity
Cost of Debt	$\text{Interest Expense} / \text{Total Debt} \times 100$
Cost of Equity	Required return on equity (CAPM estimate)
Return vs WACC	Portfolio Return - WACC
Spread (Cap - WACC)	Cap Rate - WACC

H) AR Dashboard

Route: Rentals -> AR Dashboard Frontend reference: frontend/src/pages/RentalArDashboard.tsx

KPI	Formula
Total Billed / Month	Sum of billed rent from registry (occupied units)

KPI	Formula
Collected (Latest Mo)	Sum of Rent Receivable collections for selected month
Outstanding AR	Billed - Collected (open balance)
Collection Rate	Collected / Billed x 100
Vacancy Loss / Month	Sum of rent for vacant/notice units from registry
Zero-Pay Companies	Companies with billed > 0 and collected = 0
Partial-Pay Companies	0 < Collected < Billed
Month-End Shortfall	Billed - Collected for selected month
Occupied - Billing Gap	Occupied units with vs without billing data
Top 5 by Outstanding AR	Companies ranked by highest open AR
DSO (Arrears Days)	Weighted bucket formula (same as Overview)

Collection detail table (per company x month): - Billed = registry occupied-unit rent - Collected = Rent Receivable upload (or P&L fallback)
- Outstanding = Billed - Collected - Rate = Collected / Billed x 100 - Status: Zero-Pay (0%), Partial (<100%), Paid (>=95%), Low (<85%)

I) Quick reference - core definitions

Occupancy % = Occupied Units / Total Units
Collection Rate = Collected / Billed
NOI (Ops) = Collected - Expenses
NOI (Financial) = Revenue - Expenses + Interest
Vacancy Loss = Rent from vacant units only
Arrears = Sum of (Billed - Collected), minimum 0
DSCR = Annual NOI / Annual Debt Service (EMI x 12)

J) Code locations

Area	Primary files
Operational KPIs (backend)	backend/services/rental_calculations.py, backend/routers/rentals/router.py
Overview UI	frontend/src/pages/RentalOverview.tsx
Financial KPI engine	frontend/src/utlis/rentalKpiEngine.ts
Financials KPI tab	frontend/src/pages/RentalFinancials.tsx
CFO Dashboard	frontend/src/pages/RentalCfoDashboard.tsx
CFO Portfolio	frontend/src/pages/rental/RentalCfoPortfolio.tsx
CFO data hook	frontend/src/hooks/useRentalCfoData.ts
Financial Ratios	frontend/src/pages/rental/RentalFinancialRatios.tsx
Executive Summary / PPT export	frontend/src/utlis/executiveSummaryPpt.ts, frontend/src/utlis/gatherExecutiveExportData.ts

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